

Gazal

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Education

Chitkara University, Punjab, India

Bachelor of Engineering in Computer Science Engineering

Aug 2024 – May 2028 (Expected)

CGPA: 8.45

Relevant Coursework: Operating Systems (Processes & Threads, Memory Management, File Systems, Synchronization & Concurrency), Linux/Unix Systems, Data Structures & Algorithms, Computer Networks, C/C++ Programming, DBMS, Object-Oriented Programming

Technical Skills

- **Languages:** C, C++, Java, JavaScript, Python, SQL, HTML, CSS
- **Systems:** Linux/Unix, Operating Systems, Processes & Threads, Memory Management, File Systems, Synchronization & Concurrency, Distributed Systems (Concepts), Data Structures & Algorithms, Computer Networks
- **Backend:** Node.js, Express.js, REST APIs, WebSockets, JWT Authentication
- **Databases:** MongoDB, PostgreSQL, NeonDB, Prisma ORM
- **Frontend:** React.js, Tailwind CSS, Material-UI, HTML, CSS
- **Tools:** Git, GitHub, Postman, VS Code, Vercel, Render

Projects

Talkify [Live Demo]

React.js, Node.js, Express.js, WebSockets, Prisma ORM, NeonDB

- Engineered a real-time bidirectional chat system using WebSockets, achieving low-latency message delivery across concurrent users – applying synchronization, concurrency, and distributed systems principles end-to-end.
- Designed server-side session management and connection lifecycle handling from scratch; applied systematic debugging and root cause analysis to ensure correctness across multiple concurrent clients.
- Integrated Prisma ORM with NeonDB (serverless PostgreSQL) for type-safe, scalable persistence of user data and message history; applied performance optimization for efficient, low-latency data retrieval at scale.

CareerLens [Live Demo]

React.js, Node.js, Express.js, MongoDB, JWT Authentication

- Architected a full-stack platform with RESTful APIs for PDF upload, retrieval, likes, and ratings; engineered JWT-based authentication with secure session and access control across all endpoints.
- Designed scalable MongoDB schemas with optimized indexing for efficient, high-volume data retrieval; deployed on Vercel and Render with continuous testing and validation under concurrent user load.

ReliefSync

React.js, Node.js, Express.js, MongoDB, JWT Authentication

- Built a multi-user disaster coordination platform with priority-based incident tracking; designed RESTful APIs for emergency triage, real-time status updates, and concurrent team task routing.
- Applied performance optimization and problem-solving to ensure fault-tolerant, low-latency data persistence under high-demand load; deployed for production-grade, scalable availability.

Achievements

- Solved 350+ Data Structures and Algorithms problems across competitive programming platforms, demonstrating strong algorithmic thinking and problem-solving skills.
- Cleared the Internal Smart India Hackathon (SIH) selection round, advancing among the top university teams with an innovative, solution-driven project proposal.
- Secured a Top 7 finish at Hack With Her 4.0, collaborating under time constraints to architect and deliver a full-stack technical solution addressing real-world challenges.

Experience

Outreach Executive, IEEE-CIET Student Branch

2024 – Present

Chitkara University

- Spearheaded outreach initiatives by coordinating with industry professionals and organizations to drive strategic technical collaborations and partnerships.